

Thong Minh Lai (Tom)

+44 (0)7746722114 | laiminhthong1@gmail.com

 [thong-lai](#) |  [ThongLai](#) | [Personal Website](#) | [Kaggle](#)

Manchester, United Kingdom

OBJECTIVE

Enthusiastic and skilled in AI/ML & Data Science and other technology domains, with 2 years of experience in Python, C++. I am seeking opportunities to leverage my expertise and contribute to innovative projects in AI and practical problem-solving in related fields

SKILLS

- **Programming Languages:** Python, C++/C, Java, MATLAB/Octave, JavaScript, TypeScript
- **AI/ML & Data Science:** Gradient-Boosted Trees, Outlier Detection, Deep Learning, LLM, OpenCV (YOLOv9)
- **Data Analytics:** EDA, Data Visualisation/Dashboard (Matplotlib, Seaborn, Streamlit App, Tableau)
- **ML & Data Science Libraries:** TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas
- **Cloud & Development Tools:** AWS, Metaflow, Azure, Git, Docker
- **Database Systems:** SQL, Relational Databases, Database Management Systems
- **Web Development:** React.js, Next.js, Django, Flask, Responsive Design, Web Hook, API Integration, Web/app Deployment
- **Mathematical Foundations:** Linear Algebra, Calculus, Statistics, Probabilities, Discrete Mathematics
- **Research Skills:** Academic Writing, Literature Review, Scientific Documentation
- **Soft Skills:** Leadership, Project Management, Team Collaboration, Problem Solving

EXPERIENCE

- **Awaze Vacation Rentals LTD - Europe's Leading Managed Vacation Rentals Group**  June 2025 - Present
Associate Data Scientist Manchester
 - Built Bayesian Marketing Mix Models (PyMC) to quantify channel-level ROI and model ad stock decay and saturation effects
 - Developed customer segmentation models using clustering to identify high-value cohorts for CRM retention
 - Abandoned Basket email optimisation achieving 84% recall to prioritise high-value customers for targeted outreach
 - Checkout drop-off prediction using gradient-boosted trees and SHAP/LIME to identify friction points for QA
 - Built Investment Curves dashboard (Streamlit/Snowflake) to predict incremental ROAS and optimise daily spend
 - Engaged with industry conferences including *AWS Summit 2025* and *MLOps.WTF by Fuzzy Labs*
- **Learnspot - Online Educational Platform** June 2024 - September 2024
AI Software Engineer - Summer Intern Hybrid - London
 - Designed and implemented a customer service Chatbot using Prompt Engineering with the OpenAI API
 - Integrated the Moderation API to ensure safe and appropriate interactions
 - Deployed *Chain-of-Thought* for model's reasoning with evaluation to minimise response leakage and hallucination
 - Developed an AI tutor chat feature to assist students with questions, assignments, providing real-time support

EDUCATION

- **University of Huddersfield** September 2022 - July 2025
BSc (Hons) Computer Science with Artificial Intelligence SW Huddersfield, United Kingdom
 - Overall Grade: 89.6% - Graduate Thesis: *Explainable AI (XAI) in Deep Learning Models for Credit Card Fraud Detection*
- **Ho Chi Minh University of Science - Vietnam National University** October 2020 - September 2022
BSc Information Technology Ho Chi Minh City, Vietnam
 - GPA: 3.7/4.0
 - Received Honour Scholarship for Excellent Academic Student

RESEARCH AND TECHNICAL REPORTS

T=TECHNICAL REPORT, P=PROJECT PROPOSAL

- [P.1] Lai, M. T. (2025). **Explainable AI (XAI) in Deep Learning Models for Credit Card Fraud Detection**. Graduate Thesis, University of Huddersfield. Supervisor: Prof. Hyunkook Lee
- [T.2] Lai, M. T. (2025). **Parallel Computer Architecture Clusters and Grids Benchmarking**. Technical Report, University of Huddersfield. Supervisor: Prof. Hamza Aagela
- [T.2] Lai, M. T. (2024). **Investment Allocation Analysis: A Machine Learning Approach**. Technical Report, McGill-FIAM Asset Management Hackathon

PROJECTS

• Marketing Mix Model - Awaze (Cottages.com)

April 2026 - Present

Tools: PyMC-Marketing, Robyn (Meta), Python, R, Snowflake

- Built Bayesian MMM using PyMC and to quantify channel-level ROI across PPC, TV, YouTube, and display
- Engineered ad stock decay and saturation transformations to model carryover effects and diminishing returns
- Designed upper funnel mediation models to attribute booking contribution across brand channels for Generic PPC
- Hyperparameter tuning across 20+ candidate variables

• Abandoned Basket Reminder Email Optimisation - Awaze

June 2025 - Present

Tools: LightGBM, XGBoost, SHAP, Python, Snowflake

- Developed ML classifier to predict abandoned basket recovery probability for targeted email campaigns
- Achieved 84% recall in identifying high-value customers likely to convert after email outreach
- Strategic labelling framework using timeline-based criteria and engagement patterns for imbalanced datasets
- Implemented confidence-based prediction scoring to guide differentiated CRM contact strategies

• Explainable AI (XAI) in Deep Learning Models for Credit Card Fraud Detection

October 2024 - Present

Tools: Deep Learning, SHAP, LIME, Anchors, TensorFlow, Scikit-learn

[Award] [Academic Poster] [G]

[Supervisor: Prof. Hyunkook Lee [✉](#)]

- Integrated XAI methods (SHAP, LIME, Anchors) to provide explanations for Deep Learning model predictions.
- Evaluated explainability using XAI metrics (Faithfulness, Monotonicity, and Completeness) in confidence levels.
- Compared XAI outputs across models (CNN, LSTM) to identify strengths and limitations for different user needs.
- Delivered actionable explanations to support regulatory compliance and stakeholder trust in AI-driven decisions.

• McGill-FIAM Asset Management Hackathon: Investment Allocation

September 2024 - October 2024

Tools: Machine Learning, Big Data Analytics, Financial Analysis

[Report [G](#)] [G]

- Collaborated with finance specialists as Machine Learning Specialist to develop portfolio trading strategies
- Implemented ML methods for portfolio optimisation and stock selection
- Developed models to identify key financial factors for stock performance prediction
- Conducted back-testing to validate portfolio performance using historical data
- Our strategy delivered over 7500% return from 2010 to 2024, far surpassing the S&P 500

• Eye-tracking in Reading Comprehension: Anomaly Detection

October 2023 - December 2023

Tools: Local Outlier Factor (LOF), Data Analysis, Python

[Reference [in](#)] [G]

- Developed Anomaly Detection mechanism using Local Outlier Factor for eye-tracking dataset
- Analysed 20+ features related to readers' eye movements, fixations, and directions
- Support PhD researcher Minh Huyen Nguyen to categorise fixations into "First Pass" and "Second Pass" sequences
- Improved 11.25% data analysis accuracy through systematic outlier removal and feature engineering

HONOURS AND AWARDS

• Outstanding Leadership Award

February 2023

University of Huddersfield International Office

- Led and managed the Vietnamese Society as President
- Successfully managed social media platforms achieving 2.2k+ Facebook likes and active Instagram engagement
- Recognised by International Office for impactful community leadership and cultural integration initiatives [[G](#)]

• Academic Excellence Awards

2022 - 2025

University of Huddersfield, Ho Chi Minh University of Science & High School

- "Best Overall Performance on the Final Year Project in Computing and Computer Science" Award [[G](#)]
- Merit-based Scholarship for Outstanding Academic Performance [[G](#)]
- Honour Scholarship for Excellent Academic Student at HCMUS
- Bronze Medal in Mathematics at the Excellent Student Olympic Competition

ADDITIONAL INFORMATION

Languages: English, Vietnamese, Chinese (Basic)

Technical Writing: Scientific documentation, research papers, technical blogs

Interests:

- Any AI Fields: Fraud Detection, Computer Vision, Natural Language Processing, Generative AI, Investment Analysis, Trading ML Strategies
- Cultural Exchange: International community, cross-cultural communication
- Sports: Badminton (Captain of University Badminton Team)

Personal Website: [Portfolio and Technical Blog](#)

REFERENCES

1. **Prof. Hyunkook Lee** 
Professor of Music Technology
Department of Computing and Engineering
University of Huddersfield
Email: H.Lee@hud.ac.uk
Relationship: Final Year Project Supervisor
2. **Dr. Ilias Tachmazidis** 
Senior Lecturer
Department of Computing and Engineering
University of Huddersfield
Email: I.Tachmazidis@hud.ac.uk
Relationship: Academic Advisor
3. **Dr. Muhammad Ansari** 
Senior Lecturer
Department of Computing and Engineering
University of Huddersfield
Email: m.a.ansari@hud.ac.uk
Relationship: Course Lecturer
4. **Dr. Dinh Ba Tien** 
Senior Lecturer
Faculty of Information Technology
Ho Chi Minh University of Science
Email: dbtien@fit.hcmus.edu.vn
Relationship: Former Academic Advisor

Additional references available upon request.